

# Heard but Not Heeded

## Paralinguistic Information Encoding and Loss in Audio-Language Models

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### Takeaways

- Speaking style is linearly decodable. A linear probe recovers the style (7 classes, e.g., happy, whisper) at 82–85% from the late audio encoder of all four models (chance 14.3%).
- Asked to describe the tone, the three audio-language models are right only 19.7–53.7% of the time. Qwen2-Audio’s and Chroma’s answers follow the sentence more than the style (leakage 0.98 and 0.84, where 1 = fully sentence-driven). Qwen2.5-Omni’s follow the style more (0.27).
- Why Omni differs is not settled. Our leading hypothesis is its speech-generation objective, but it also uses an MLP projector rather than a linear one. Retraining only Qwen2-Audio’s linear projector with gradient reversal did not change leakage (0.977 → 0.979).



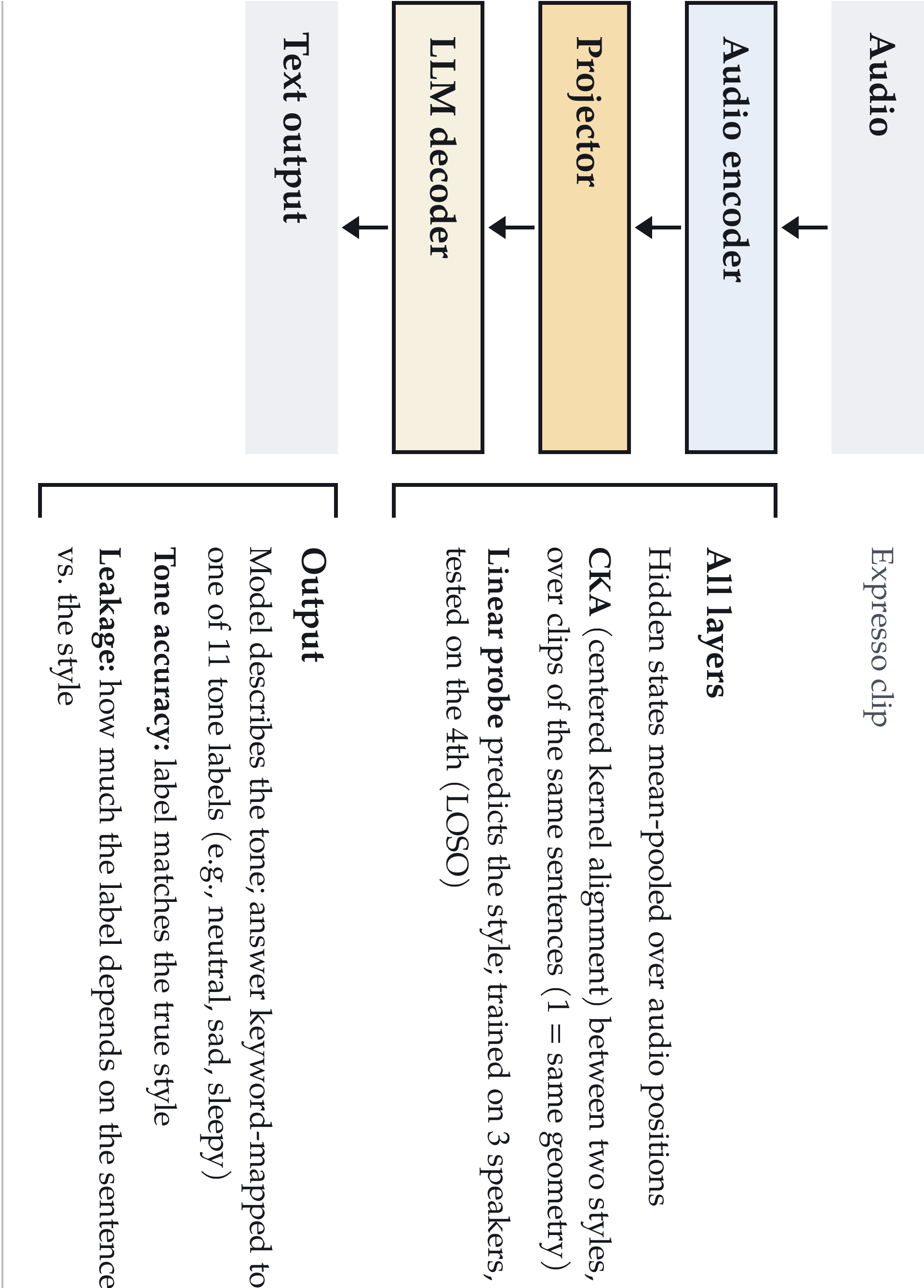
## 1 Motivation

Audio-language models (ALMs) sometimes appear insensitive to tone. In three open ALMs, plus Whisper as a reference, we ask whether paralinguistic information (how something is said; here, speaking style) is absent from their internal representations, or present but not reflected in their answers.

## 2 Setup

Expresso (Nguyen et al., 2023): 4 speakers read the same sentences in 7 styles (confused, default, enunciated, happy, laughing, sad, whisper), 10,636 clips in total. We pair clips with the same sentence and speaker but different styles, giving 31,896 pairs.

## 3 Methods

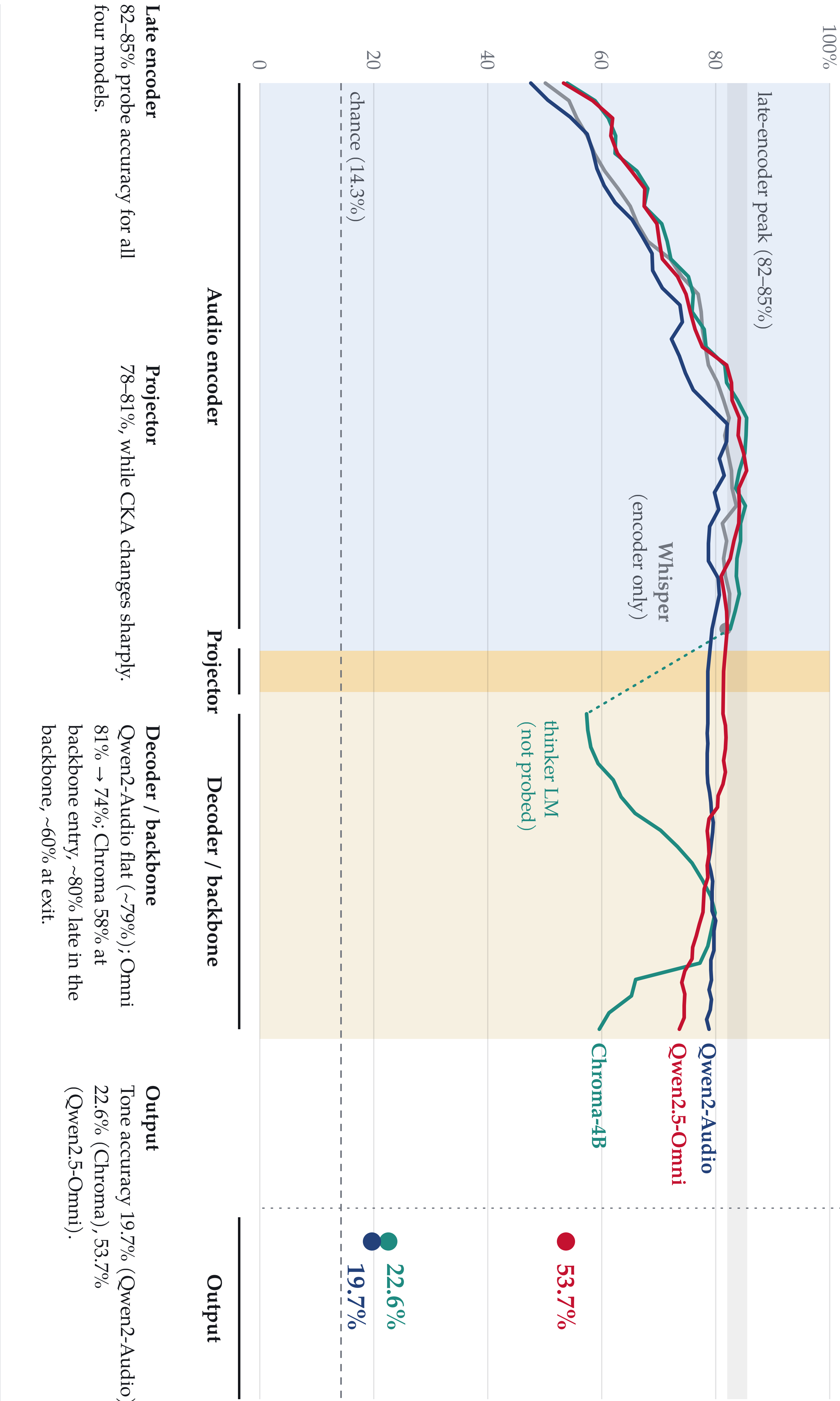


| Model                   | Layers probed                                            | Objective                                  |
|-------------------------|----------------------------------------------------------|--------------------------------------------|
| Whisper-large-v2        | 32 encoder                                               | Speech recognition                         |
| Qwen2-Audio-7B-Instruct | 33 encoder → linear projector → 33 LLM                   | Text generation                            |
| Qwen2.5-Omni-7B         | 32 encoder → MLP projector → 28 LLM                      | Text + speech generation                   |
| Chroma-4B               | 32 audio encoder → 16 speech backbone → 4 speech decoder | Text (thinker) + speech with voice cloning |

The projector maps audio-encoder outputs into the LLM’s input space. Whisper gives no tone answer. Chroma’s language model (“thinker”, 36 layers) sits between its audio encoder and speech backbone and is not probed. Its CKA across styles is 0.98–1.0.

## 4 Encoder–output gap

**Lines:** linear probe accuracy (7-way, LOSO) at each layer, with stages aligned across models. **Dots:** tone accuracy of the model’s text output. The probe is trained and the tone answer is zero-shot, so the two are not directly comparable.



**Late encoder**  
82–85% probe accuracy for all four models.

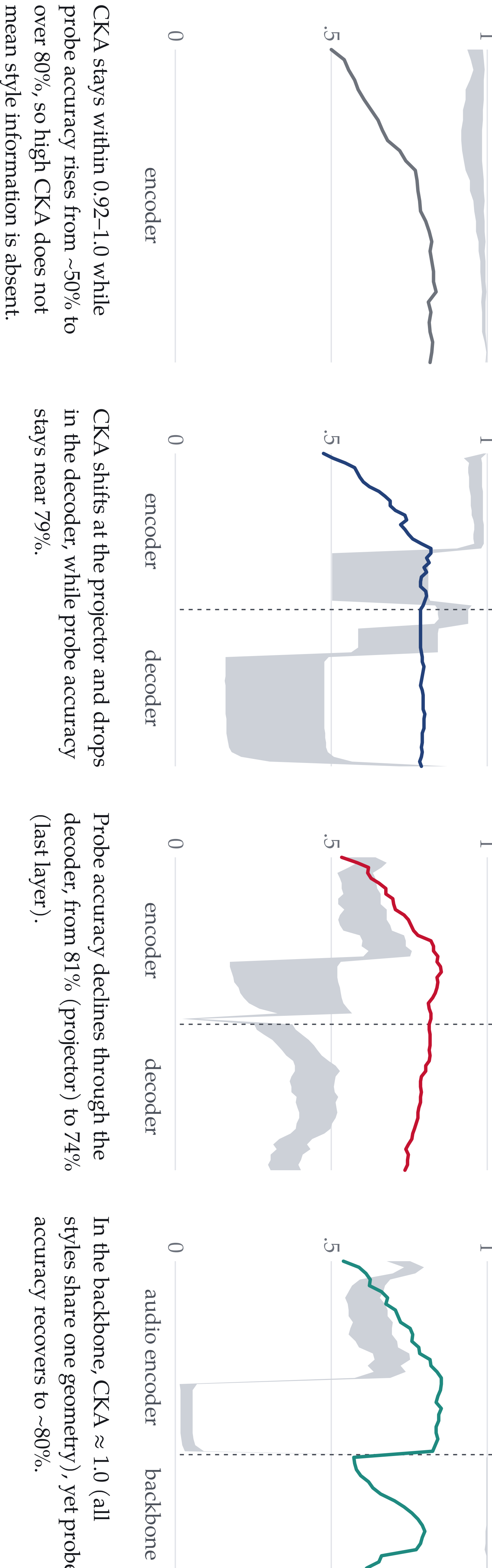
**Projector**  
78–81%, while CKA changes sharply.

**Decoder / backbone**  
Qwen2-Audio flat (~79%); Omni 81% → 74%; Chroma 58% at backbone entry, ~80% late in the backbone, ~60% at exit.

**Output**  
Tone accuracy 19.7% (Qwen2-Audio), 22.6% (Chroma), 53.7% (Qwen2.5-Omni).

### CCA vs. linear probe

— linear probe accuracy (0–1)    ■ CKA, min–max over all 21 pairs of the 7 styles    | stage boundary



## 5 Output-level behavior

Prompt: “Reply in English only. What is the person saying? Describe the tone briefly (e.g., neutral, happy, sad, angry, whispering, confused, laughing).” Leakage uses normalized mutual information (NMI) over all clips to compare how much the tone label depends on which sentence was spoken vs. how it was spoken.

| Model        | $leakage = \frac{NMI(label, sentence) + NMI(label, style)}{NMI(label, sentence) + NMI(label, style)}$ |                   |         |  | Same label in all 7 styles |
|--------------|-------------------------------------------------------------------------------------------------------|-------------------|---------|--|----------------------------|
|              | NMI(label, sentence)                                                                                  | NMI(label, style) | Leakage |  |                            |
| Qwen2-Audio  | 0.234                                                                                                 | 0.006             | 0.977   |  | 30.5%                      |
| Qwen2.5-Omni | 0.140                                                                                                 | 0.374             | 0.272   |  | 0.6%                       |
| Chroma-4B    | 0.254                                                                                                 | 0.047             | 0.844   |  | 19.9%                      |

Leakage is near 1 when the label tracks the sentence and near 0 when it tracks the style. The last column counts sentences whose label is the same in every style.

### Per-style tone accuracy



Styles are sorted by Qwen2.5-Omni’s accuracy. Gray lines join the three models in each row.

### Example outputs

| Model        | “Monday there’s gonna be haze but Tuesday look for thunderstorms” |            |          | “Nonday there’s gonna be haze but Tuesday look for thunderstorms” |  |  |
|--------------|-------------------------------------------------------------------|------------|----------|-------------------------------------------------------------------|--|--|
|              | sad                                                               | whisper    | laughing | confused                                                          |  |  |
| Qwen2-Audio  | neutral                                                           | neutral    | neutral  | neutral                                                           |  |  |
| Qwen2.5-Omni | sleepy                                                            | whispering | happy    | neutral                                                           |  |  |
| Chroma-4B    | neutral                                                           | neutral    | neutral  | neutral                                                           |  |  |

Tone label each model gives for the same sentence in four spoken styles. Omni labels sad speech “sleepy”, which we count as incorrect. It may reflect a mismatch in tone vocabulary.

## 6 Discussion

- Qwen2-Audio and Qwen2.5-Omni have similar encoders but leakage of 0.98 vs. 0.27. Omni is also trained for speech generation and uses an MLP projector. With two models, we cannot separate these factors.
- Chroma’s backbone receives inputs with CKA ≈ 1.0 across styles, yet its probe accuracy recovers to ~80%. This suggests the backbone partly re-encodes style from context.

### Retraining the projector

We retrain only Qwen2-Audio’s linear projector (1280-dim → 4096-dim, ~5M params) with a style-classification head and an adversarial content head that predicts the sentence (20 text clusters) through a gradient reversal layer (GRL), to keep style decodable while suppressing content.

| Leakage ratio        | Baseline | GRL   |
|----------------------|----------|-------|
|                      | 0.977    | 0.979 |
| NMI(label, style)    | 0.006    | 0.004 |
| NMI(label, sentence) | 0.234    | 0.203 |
| Tone accuracy        | 19.7%    | 18.0% |

The style head reached 93.3% validation accuracy by epoch 4. The checkpoint we evaluated (epoch 2, weakest content head) left leakage unchanged.

### Limitations

- One dataset (Expresso) with only 4 speakers, so results may not generalize to larger speaker populations
- One prompt, keyword-mapped to tone labels
- Mean pooling removes temporal dynamics (e.g., laughing)
- Probe accuracy shows that style is decodable, which does not mean the model uses it



Paper  
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